Continuous Distillation

Group #2
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Introduction

- The objective is to utilize a continuous distillation column containing an ethanol/water mixture to investigate the effects of varying feed position on continuous operation.
- From this experiment, a McCabe-Thiele analysis will be performed to calculate the overall column efficiency to compare with the efficiency of the concentration from the bottom and top compositions

Theory

- Distillation Column

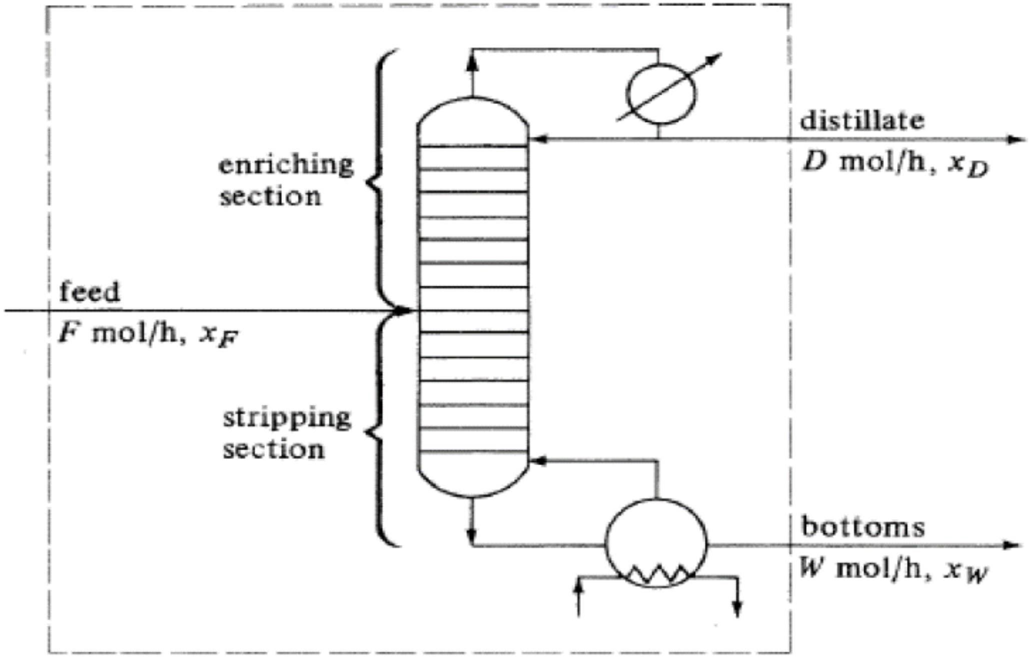

Figure 1 – Distillation Column[1]

Theory (Cont.)

- McCabe-Thiele Diagram

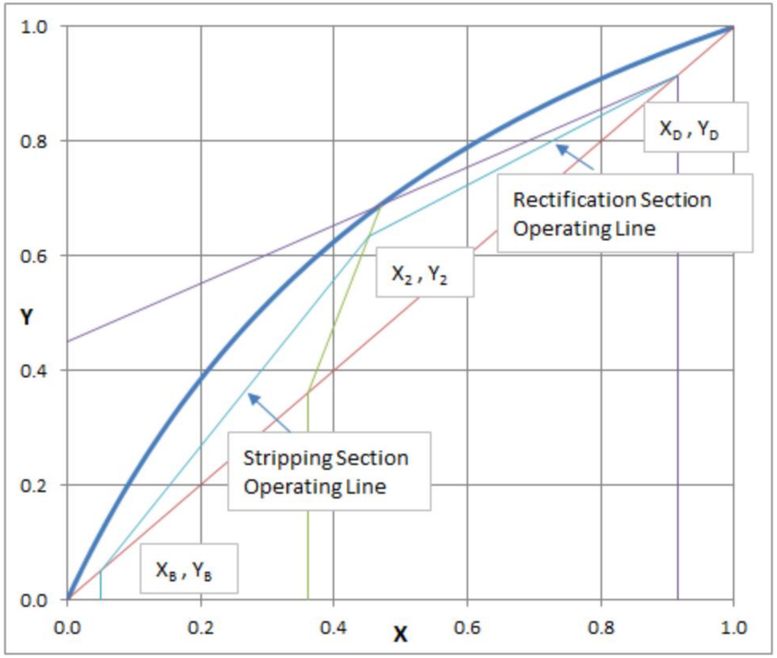

Figure 2 – McCabe-Thiele Diagram[3]

Theory (Cont.)

- The equations used for experiment:

❑ Fenske Equation:

$$N_{min} = \frac{log\left[\left(\frac{x_d}{1-x_d}\right)\left(\frac{1-x_b}{x_b}\right)\right]}{\log(\alpha_{avg})}$$

- Where:
- $N_{min} = Minimum\ number\ of\ plates\ at\ total\ reflux$
- $x_d = Mole\ fraction\ of\ distillate$
- $x_b = Mole\ fraction\ of\ bottoms$
- $\alpha_{avg} = \sqrt{\alpha_t \alpha_b} =$
  $Average\ relative\ volatility\ \ of\ LK\ to\ HK\ at\ top\ and\ bottom$

Theory (Cont.)

- Vapor Liquid Equilibrium (VLE) Volatility:

$$\alpha_{EtOH} = \frac{\frac{y_{EtOH}}{x_{EtOH}}}{\frac{1 - y_{EtOH}}{1 - x_{EtOH}}}$$

- Where:
- $y_{EtOH} = Liquid\ phase\ mole\ fraction\ of\ ethanol$
- $x_{EtOH} = Vapor\ phase\ mole\ fraction\ of\ ethanol$

Theory (Cont.)

Top section overall material balance:

$$V_n = L_{n+1} + D + F$$

$$y_n V_n = x_{n+1} L_{n+1} + x_D D + x_F F$$

- Where:
- $V_n = Flowrate\ of\ vapor\ at\ stage\ n$
- $L_{n+1} = Flowrate\ of\ liquid\ at\ stage\ n+1$
- $D = Flowrate\ of\ distillate\ product$
- $F = Flowrate\ of\ feed$
- $y_n = Vapor\ composition\ at\ stage\ n$
- $x_{n+1} = Liquid\ composition\ at\ stage\ n+1$
- $x_D = Composition\ in\ distillate\ stream\ D$
- $x_F = Composition\ in\ feed\ stream\ F$

Theory (Cont.)

- **Bottom section overall material balance:**

$$V_n = L_{n+1} - B$$
$$y_n V_n = x_{n+1} L_{n+1} - x_B B$$

- Where:
- $V_n = Flowrate\ of\ vapor\ at\ stage\ n$
- $L_{n+1} = Flowrate\ of\ liquid\ at\ stage\ n+1$
- $B = Flowrate\ of\ bottoms\ product$
- $y_n = Vapor\ composition\ at\ stage\ n$
- $x_{n+1} = Liquid\ composition\ at\ stage\ n+1$
- $x_B = Composition\ in\ product\ stream\ B$

Theory (Cont.)

- Rectifying Line Function:

$$y_{n+1} = \left(\frac{L}{V}\right)x_n + \left(1 - \frac{L}{V}\right)x_D$$

- Stripping Line Function:

$$y_{n+1} = \left(\frac{L}{V}\right)x_n + \left(\frac{L}{V} - 1\right)x_B$$

- Efficiency:

$$Efficiency = \frac{Number\ of\ theoretical\ plates}{Number\ of\ actual\ plates} \times 100\%$$

Apparatus

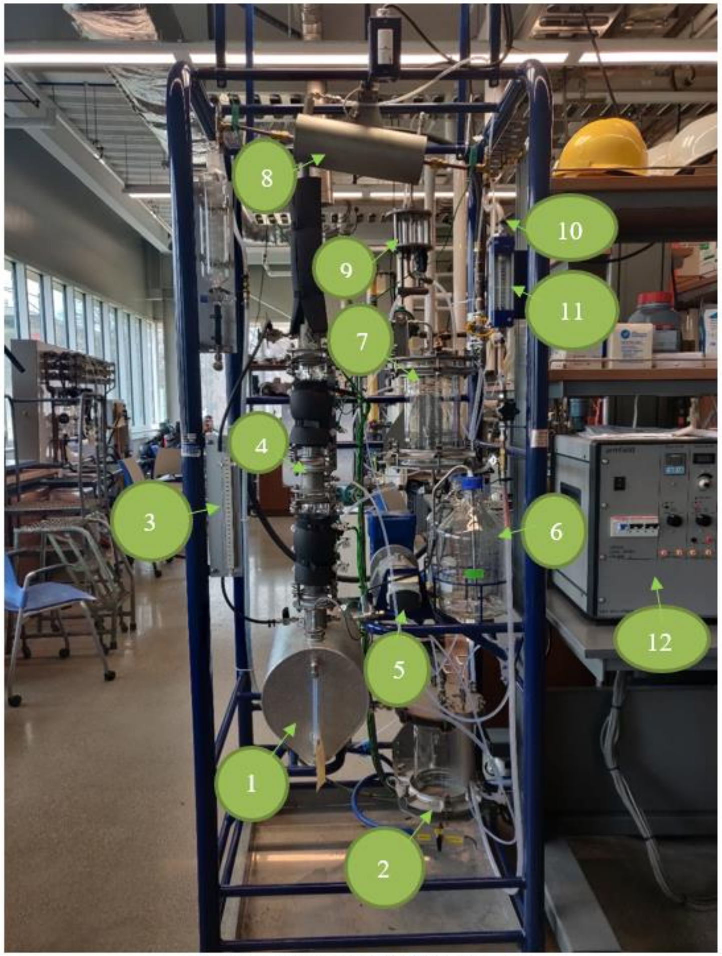

Figure :1 Continuous Distillation Column

Number	Apparatus	Description
1	Reboiler	Works to heat up the feed
2	Bottom Product Collector	Where the bottom sample is collected, or purged
3	Manometer	Measures pressure
4	Distillation Column	Where the water and ethanol are separated through trays
5	Feed Pump	Pumps the mixture into the column
6	Feed Tank	Where the feed mixture is contained
7	Top Product Collector	Where the top sample is collected
8	Condenser	Cools vapor into liquid from the top of the column
9	Decanter	Where sample from distillate is collected
10	Cooling Water Supply	Where water flows through condenser
11	Rotameter	Measures flow rate of cooling water
12	Armfield Control Console	Used to control the column: set up reflux ratios, change the temperature, power the reboiler, and control pump speed.

Material & Supplies

Materials/Supplies	Brief Description
Ethanol	Solution added into the feed and into the reboiler
Water	Solution added into the feed and into the reboiler
Water Bath	Used to cool down the samples in the test tubes
Test Tubes	Used to collect the samples
Graduated Cylinder	Used to collect the water sample for the calibration graph
Stopwatch	Used to measure the time intervals
Refractometer	Used to measure the refractive index
Pipette	Used to add sample onto the refractometer
Parafilm	To cover the sample in the test tubes

Procedure

- Experiment A:
  - Add water into the feed tank
  - Change the speed of the pump and collect a volume of 30mL for each speed
  - Take a stopwatch and record the time it takes to collect that amount of volume at that speed

Procedure Continued…

- Experiment B:
  - Make 10L of a 10wt% ethanol/water mixture and 3L of a 30wt% ethanol/water mixture
  - Set the equipment at total reflux by turning off the reflux ratio timer on the console
  - Connect the clear feed tubing from the pump outlet to the column between the 4th and 5th tray and turn on the power of the control
  - Turn on the power of the control panel and set the temperature selector to T9

Procedure Continued

- Set the flowrate on F11 to 3L/min and turn the power controller for the reboiler clockwise until it reaches 1.5kW
- Switch on the temperature selector on for T8, T7, T6, T5, T4, T3, T2, and T1
- Once the distillate builds up and steadys out, turn the reboiler power to 0.75 kW
- Start the experiment with total reflux and switch on the reflux ratio to 5:1 and the feed flow to a pump speed of 15%.

Procedure Continued

  - Take 3 samples at a time interval of 5 minutes apart and record the temperatures of the column and the refractive index
- Experiment C:
  - Change the feed to the top plate
  - Start the experiment with total reflux and switch on the reflux ratio to 5:1 and the feed flow to a pump speed of 15%.
  - Take 3 samples at a time interval of 5 minutes apart and record the temperatures of the column and the refractive index

Experimental Challenges

- Make sure to take the samples simultaneously
- Make sure to measure 5 minutes exactly
- Make sure to purge before collecting the samples
- Parafilm samples right away
- Add the ethanol last into the mixture, otherwise the ethanol will evaporate

Safety

- Be careful when taking samples from the reflux drum as the sample is coming out fast and is extremely hot